

Q.30 Explain the terms briefly. (CO7)

- a) Frequency b) mean
- c) median d) mode
- e) standard deviation

Q.31 Explain different types of Inspection. (CO1)

Q.32 Explain the term Interchangeability in connection with measuring instruments. (CO2)

Q.33 Write a short note on KAIZEN. (CO6)

Q.34 Explain different types of Gauges with diagram. (CO2)

Q.35 Explain clinometers with its uses. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the construction and working principle of micrometer with help of neat sketch. (CO2)

Q.37 What are variable charts? Explain them with neat sketch diagram. (CO5)

Q.38 Explain concept of TQM model. Also state its various advantages and functions. (Co6)

No. of Printed Pages : 4

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6th Sem / Mech, T&D, Prod, Mecatronics, GE, CNC, CAD/CAM, Found. & Forg., Mech. Engg. (Fabrication Tech.), Mech (CAD/CAM Design & Robotics)

Subject:- Inspection and Quality Control / Metrology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 One Micron = (CO1)

- a) 10^{-2} m b) 10^{-4} m
- c) 10^{-6} m d) 10^{-8} m

Q.2 Write the full form of S.Q.C? (CO6)

- a) Statistical Quality Control
- b) Systematic Quality Control
- c) Sequential Quality Control
- d) None

Q.3 Least count of vernier calliper is (CO2)

- a) 0.01 mm b) 0.04 mm
- c) 0.02 mm d) 0.05mm

Q.4 Sine bar is specified by (CO2)

- a) Distance between its two rollers
- b) Size of the rollers
- c) Total length of its flat surface
- d) Thickness of its flat portion

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(4) 181762/171762/121762
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Q.5 Which of the following is control chart for variable? (CO2)

- a) $\bar{X}$ chart b) C chart
c) U chart d) All of the above

Q.6 Interchangeability helps in (CO1)

- a) Increasing cost b) Reducing cost
c) Does not affect cost d) None of these

Q.7 tolerance on a component are specified (CO3)

- a) To obtain desired fits
b) Because it's not possible to manufacture required size
c) to obtain high accuracy
d) to have allowance

Q.8 Module of a gear is (CO2)

- a) D/T b) T/D
c) 1/D d) None of these

Q.9 A gear is specified by (CO2)

- a) Tooth profile (T) b) Module
c) Pressure angle d) All of these

Q.10 One inch is (CO1)

- a) 25.4mm b) 24.5 mm
c) 26 mm d) 23.5 mm

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 ISO stands for _____ (CO6)

Q.12 The agency to frame national standards in India is _____ (CO6)

(2) 181762/171762/121762
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Q.13 Define centralized Inspection. (CO2)

Q.14 Least count of micrometer is _____ mm (CO2)

Q.15 Define Range. (CO2)

Q.16 Expand TQM (CO3)

Q.17 What is the other name of P-chart? (CO3)

Q.18 Define circular pitch. (CO3)

Q.19 Define Interchangeability. (CO1)

Q.20 Define Random Error. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Define flatness. Describe any one method of testing flatness. (CO3)

Q.22 Explain the construction of LVDT with neat sketch. (CO7)

Q.23 Define Transducer. Classify transducers on the basis of output. (CO7)

Q.24 What is FIT? Explain its types. (CO2)

Q.25 Write any five differences between variable and attribute chart. (CO5)

Q.26 Explain sine bar with neat sketch. (CO3)

Q.27 Define error. Explain any two types of error. (CO4)

Q.28 Discuss double sampling plan. (CO5)

Q.29 Write the characteristics of Normal distribution. (CO4)

(3) 181762/171762/121762
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